Chinmay Govind

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Education

University of Pennsylvania, School of Engineering and Applied Sciences Aug. 2024 – May 2028
B.S.E in Artificial Intelligence - GPA: 3.99 | M.S.E in Robotics - GPA: 4.00 *Philadelphia, PA*
- Coursework: Operating Systems, Signal Processing, Stochastic Processes, Machine Perception, NLP
- Honors: Wharton Datathon 1st Prize, SIG Poker Showdown 2nd Place, PURM Research Award

Experience

Susquehanna International Group June 2027 – Aug. 2027
Incoming Quantitative Trading Intern *Philadelphia, PA*

Shopify June 2026 – Aug. 2026
Machine Learning Engineer Intern, Search Team *New York, NY*
- Developed a novel metric for Shop App search quality that flags high-quality products ranked too low in results, validated across 12M+ search ratings.
- Fixed a URL-matching bug in the search quality scraper with SQL regex, boosting match rate by 5%.

Penn Electric Racing (Formula SAE) Sept. 2024 – Present
Software Lead *Philadelphia, PA*
- Lead a 20-person software team building firmware, telemetry, and data infrastructure for the team's first four-wheel-drive electric racecar.
- Brought up STM32 C++ firmware for the power distribution and powertrain control boards, including sensor calibration, BSPD fault monitoring, and writing drivers for new motor controllers.

UPenn Department of Computer Science Dec. 2024 – Present
Head Teaching Assistant - CIS 1600 (Discrete Math) + ESE 5420 (Data Science) *Philadelphia, PA*
- Manage staff of 30 TAs and 500+ students, overseeing course logistics, grading, and exams.
- Developed automatic grading systems for automated feedback and academic integrity policy verification, saving 20+ TA hours/week and $6,000 per semester.

Penn Whales Research Group May 2025 – Jan 2026
Machine Learning Researcher *Philadelphia, PA*
- Engineered a machine learning pipeline to locate whales in 3D space from underwater acoustic recordings.
- Generated 50GB of simulated acoustic data for model training and developed custom Python CUDA-optimized training scripts, cutting training time by 70%.

Projects

Car Data Server | *Rust, WebAssembly, Python, PostgreSQL, S3, React* Jan. 2026 – Present
- Built season-wide telemetry database of 8B+ data points over 1,625 sessions, queryable via SQL, Python and MCP.
- Automated car-to-cloud log backup, decoding, upload and tagging, recovering 1,395 logs and managing 3,000+.
- Ported Rust telemetry dashboard to web via WebAssembly, adding mobile support, scripting and offline replay.

cgovind.com | *Python, Flask, WebSockets, SQLite, Nginx, AWS EC2, GitHub Actions* 2026 – Present
- Built and deployed a personal web platform hosting 4 real-time multiplayer games, a poker trainer, and site-wide chat with shared accounts, serving 228K+ page views from 6,000+ unique visitors.
- Developed a poker trainer that simulates player behavior and scores each decision with Bayesian inference.
- Created a multiplayer 3D racing game with ghost replays and server-side anti-cheat validation of submitted runs.

Prediction Market Bot | *Python, Bash Scripting, Linux* Sept. 2026 – Present
- Built a trading bot for election prediction markets, sniping mispriced orders and arbitraging races.
- Priced 110+ markets off Kalshi after lead-lag analysis and backtests, returning +9.2% in two weeks of live trading.

Technical Skills

Languages: Java, Python, JavaScript, C, C++, Rust, PHP, TypeScript, Go, SQL, .NET, LaTeX, OCaml, C#, Bash
Frameworks/Tools: Git, Github, Docker, Kubernetes, Node.js, MongoDB, PostgreSQL, Spring Boot, Flask, Django, Apache, React, Selenium, Postman, GDB, Ghidra, Arduino, Jupyter Notebook, Linux, UNIX, PowerShell
Libraries/Platforms: PyTorch, NumPy, Scikit-Learn, Pandas, Matplotlib, MATLAB, CUDA, OpenCV, TensorFlow, Keras, AWS Lambda, AWS S3, GCP Vertex AI, Firebase, WebRTC, Nginx, FastAPI, Redis